Outlook

Re: Update on Multilab RRR

From Chris Isloi <barbu.chris@gmail.com>
Date Tue 02-Jul-24 4:10 PM
To Adam Wang <adam.wang@jcu.edu.au>
Cc Kaiqin Chan <Kaiqin.chan@jcu.edu.au>

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Sure, thanks. Just putting it out there.

On Tue, Jul 2, 2024 at 11:02 AM Adam Wang <adam.wang@jcu.edu.au> wrote:

> Hi Chris
>
> We're still waiting for some labs to send their data, so there will be some more delays before we get i touch again regarding final author lists etc
>
> Cheers
> Adam
>
> ---
>
> **From:** Chris Isloi <barbu.chris@gmail.com>
> **Sent:** Tuesday, July 2, 2024 3:59 PM
> **To:** Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
> **Cc:** Adam Wang <adam.wang@jcu.edu.au>
> **Subject:** Re: Update on Multilab RRR
>
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>
> Hi, Adam
>
> quick question. Is the data we sent OK?
>
> Also, will there be an authorship list we can complete? A few of our collaborators are not listed anywhere.
>
> Thanks.
>
> Chris.
>
> On Tue, Jun 18, 2024 at 4:16 PM Chris Isloi <barbu.chris@gmail.com> wrote:
>
> > Dear Kai,

For Study 1, please find attached the dataset with preliminary ratings from the 2 coders and their final agreement for each target-non-target low1,2/high1,2 items. We computed Cohen's Kappa for each pair and the overall average across items, and attached the code we used (RMarkdown and the .html output).

For Study 2, please find attached the dataset with preliminary ratings from the 2 coders and their final agreement for each item. For this study, we did not proceed with an ICC computation. While we could do that with Mattias Gamer's "irr" package, for instance, we believe this is wrong, since coder ratings represent count, not continuous data. We would highly recommend using a generalized mixed model (Poisson distribution) with subjects and raters as random intercepts.

Best,

Chris

On Wed, Apr 10, 2024 at 5:44 PM Kaiqin Chan <Kaiqin.chan@jcu.edu.au> wrote:

Dear Chris,

<u>For Study 1</u>
My sentiments are the same as yours. Kappa must be computed pairwise and an average of Kappas will be taken; there is just no other way to compute Kappa. An alternative is to compute a traditional ICC, by comparing the sum of coder 1's data against coder 2's data. Each carries its own assumptions.

You are right that the original paper did not report reliability. They may have assumed that a correct answer would have been unambiguous (participants either said that word, or they didn't), and if the assumption was true, then interrater reliability was unnecessary. But we were being a bit more conservative because we weren't sure if this assumption was necessarily true. From the data that we have already received, it wasn't always the case that the initial interrater reliability was 100% (but it was generally high), so it seems that the original authors' assumptions may not have been true.

I do agree that the fact that coders 1 and 2 have to reach a consensus eventually makes an initial calculation of kappa (or ICC, if you wish) irrelevant. But there are two reasons to keep it:

1. We preregistered it;
2. We will know to what extent the original authors' assumptions about reliability were correct or not.

I don't think you need to fret over the calculation of kappa (or ICC). At the end of the day, we will still calculate these for all labs based on the data they submit.

<u>For Study 2</u>
Kappa won't be the correct analysis cos the responses are not binary; ICC is the only way, as you rightly pointed out. It is imperfect because, as you rightly pointed, it assumes that individual responses are normally distributed. But I am not aware of alternatives. Personally, I'm not too worried about it. Many writers about normality assumptions do not distinguish normality of construct vs. normality of constituent items. In traditional multi-item scales, individual items rarely are normally distributed, but when averaged, the construct can be quite normally distributed. There is no presumption that constituent items of a construct need to be normally distributed. (I discovered this when I analyzed a Machiavellianism questionnaire.)

Eventually we will used mixed effects (the Editor requested for it), but I don't think mixed effects solves this problem. As far as I am aware, mixed effects does not deal with situations regarding the dependent variable reliability.

Unlike Study 1, ICC is important in Study 2. But again, do not fret over it if you cannot compute it. You can still send us your data even if your ICC is unknown. If we find it too low, one possibility is that we request you to revisit your coding.

Some labs did inform us about their computed reliabilities when they sent us their data. But they did not send us a syntax or anything, hence there is nothing for us to share with you.

Regards,
Kai Qin

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Wednesday, April 10, 2024 9:36 PM
To: Adam Wang <adam.wang@jcu.edu.au>; Raluca Diana Szekely <ralucaszekely@gmail.com>
Cc: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Hi Kai,
Hi Adam,

cc-ing our collaborator Raluca, whom I've been going back and forth over kappa/ICC.

For Study 1 - There's no method that I'm aware of to come up with a kappa metric computed directly from the percentage of cells that the 2 coders are consistent on. Kai did kappas on "pairs of variables" which, in my understanding means the final kappa metric will be an average of all those pairs. Is that correct?

For Study 2 - Things get somewhat more complicated here. For one, as far as I'm aware, we can't compute kappas on count data, so on this one we'll have to run an ICC. A few weeks back, Adam suggested we just go with the default ICC in SPSS, which also has a problem. Given that the scoring for each item requires the rater to count (add what), this violates assumptions of the linear model (e.g., normality). I am not sure if SPSS handles this issue. **Specifically, one would need to run a generalized mixed effects model with participant and items as random intercepts, as well as a poisson link.** Any ideas?

This is less of an issue with the idea that it's a direct replication, like Ian Hussey had brought up regarding the narrow/broad scoring strategies. The original 1987 paper from Study 2 just mentions a .94 reliability between two coders that was averaged over the three measures, while the 1992 paper from Study 1 doesn't mention inter-rater reliability. The issue we have is that inter-team variability in running these metrics might result in heterogenous results. I wonder if all teams finished coding their kappas/ICCs and if we could take a look at the way they did it.

Thanks.

Chris.

On Fri, Apr 5, 2024 at 4:23 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

> My understanding is that the single value is just a reflection of the percentage of cells that the 2 coders are consistent on, so ifall cells are equal, you get 1, and if all cells are different, you get 0. @Kaiqin Chan Kaiqin is that the correct understanding?
>
> Adam
>
> ---

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Friday, April 5, 2024 9:18 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Cc: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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It now makes sense that it's not supposed to involve the final columns, but is still doesn't explain how you end up with just one ICC metric when you compare many sets of columns. Don't you get multiple ICC (kappa) values? This goes back to my question about Kai saying he had compared "pairs" of variables.

On Fri, Apr 5, 2024 at 4:14 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

Hi Chris

The point of this ICC calculation is to see how similar the coders are, so it is not supposed to involve the final columns. The comparison should be between the 2 coders on each trial, so for example, these 2 screenshots should be compared directly to each other:

target.low1.coder1	target.low2.coder1	target.high1.coder1	target.high2.coder1
1	1	0	1
0	0	0	1
1	0	0	1
0	0	1	0

target.low1.coder2	target.low2.coder2	target.high1.coder2	target.high2.coder2
0	1	1	1
0	0	0	0
1	1	0	1
1	1	1	1

Does that make sense?
Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Friday, April 5, 2024 9:08 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Cc: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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The problem is that there isn't just one coder1 column and one coder2 column, but many of them.

target.low1.coder1 target.low2.coder1 - both in blue. That's one possible metric.

target.high1.coder1 target.high2.coder1 - both in yellow. Another possible metric.

My initial question was what are the pairs Kai/we would run the metric on, if there's no "final" column completed as of yet.

On Fri, Apr 5, 2024 at 3:59 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

Hmm, perhaps Kaiqin might explain this better

@Kaiqin Chan Kaiqin are they able to get the Kappa value from the coder1 and coder 2 columns? Or am I missing something here?

Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Friday, April 5, 2024 8:55 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Cc: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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I'm sure it's mostly on my part, having never run an ICC/Cohen's kappa analysis. I see them all the time in papers and am just now realizing that they're somewhat more complex then you would think seeing a unique value. Therefore, my apologies.

*"we mainly want to look at the final values after **all** disagreements are reconciled"* - again, wouldn't these be 1, if all disagreements are reconciled? How can you have variation in an inter-rate metric if every disagreement is reconciled?

"in the unlikely event that the coders begin with very different coding outcomes, you can get the initial ICC value from their initial coding" - still unsure from which columns you get the initial kappa metric. As I understand it now, they can't complete the "final" column categories until they reach an agreement. Therefore, running any inter-rater reliability metric on those columns would already put it at perfect reliability aka 1.

On Fri, Apr 5, 2024 at 3:32 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

Hi Chris

Sorry for the confusion, the wording in the manuscript is indeed not the best.

I think what we were trying to say is that the initial ICC values are not that important, as we mainly want to look at the final values after all disagreements are reconciled. The bit about the initial value being less than ideal is really meant to convey the idea that is the 2 coders are having very different approaches, then they need to get on the same page somehow. That is, in the unlikely event that the coders begin with very different coding outcomes, you can get the initial ICC value from their initial coding, and then meet with them to resolve any discrepancies in understanding, and then ask them to resolve and disagreements accordingly. This is why there are three column categories (coder1, coder 2, and final).

Hope this explains

Cheers
Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Friday, April 5, 2024 8:19 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Cc: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Right, but then on which columns would the initial Kappa be run on? I saw Kai mention he ran them on pairs of variables (I'm guessing A_period.1.coder1 with A_period.1.coder2 and so on?), but the manuscript only mentions one Kappa metric. Where do you get the initial unique metric from?

On Fri, Apr 5, 2024 at 3:14 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

Hi Chris

I think what he meant was the eventual agreement, whereas we want the Kappa value for initial levels of agreement

Cheers
Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Friday, April 5, 2024 8:12 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Cc: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Hi Adam, hi Kai,

sorry to drill so much over this, but wouldn't Kai' instruction that coders will "need to agree on every value" put every Kappa at 1 (perfect agreement)?

On Wed, Mar 27, 2024 at 6:55 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

Yes it is needed otherwise we wont get an inter rater reliability metric.

Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Thursday, March 28, 2024 12:18 AM
To: Adam Wang <adam.wang@jcu.edu.au>
Cc: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Thanks, Adam.

This still doesn't clarify whether a cohen's kappa is still warranted, as per Kai's question.

On Wed, Mar 27, 2024 at 5:20 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

> Hi Kaiqin and Chris
>
> According to my briefing session and the relevant section in the predata manuscript: it is possible that the initial inter-rater agreement levels from some labs were less than “excellent” (i.e., ICC < .75; Fleiss, 1986). Participating researchers have been asked to flag such unlikely instances as early as possible should they arise. In response, the lead researcher team would engage in discussions with these labs in order to identify and correct any issues, such as instruction interpretation discrepancies or individual differences in pronunciation perceptions. Upon eventual reconciliation, data from these labs would still be included in the primary analyses although exploratory analyses may be conducted to see if the exclusion of these labs would significantly alter the results of the primary analyses.
>
> So the next steps are fairly straightforward - I will note down that Chris' lab had an initial inter-rater reliability that was less than optimal, and Chris' experimenters get together to agree on one set of results, and then send us the final results. Hope this clarifies!
>
> Cheers
>
> Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Wednesday, March 27, 2024 9:53 PM
To: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Cc: Adam Wang <adam.wang@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Thanks, Kai.

Any chance you could also merge the second database, so we have something unitary to work on? Thanks.

Chris.

On Wed, Mar 27, 2024 at 3:49 PM Kaiqin Chan <Kaiqin.chan@jcu.edu.au> wrote:

Dear Chris,

The attached Excel sheet is your combined Study 1 data.

If coders were rating using likert scales, then the interrater reliability would be computed and if it's decent, then an average score of both coders would be taken as the final DV.

In the present study, the ratings are binary. If kappa is high, we can't take an average of two binary values as the final DV, because some DVs will end up being 0.5, which is an meaningless value. Hence for binary variables, coders will necessarily have to agree whether any discrepancy would be 1 or 0, regardless of how high or low the kappa is. To some extent, I agree that the kappa isn't very meaningful because at the day, coders will still need to agree on every value, thus making any initial kappa quite pointless. I actually didn't realize this until you raised this.

Adam, do you agree? Is computing Cohen's kappa still necessary? In our RRR manuscript, we did say we were going to compute an ICC, which we probably should have said "Cohen's kappa". But Chris also raised a valid concern about the utility of computing kappa in the first place.

Kai Qin

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Wednesday, March 27, 2024 8:44 PM
To: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Cc: Adam Wang <adam.wang@jcu.edu.au>
Subject: Re: Update on Multilab RRR

Also, one more question. Wouldn't the consensus columns come after calculating kappa? Isn't that the point? Not sure how to proceed.

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On Tue, Mar 26, 2024 at 7:06 PM Chris Isloi <barbu.chris@gmail.com> wrote:

Hi Kai,

not sure if you managed to merge the databases for each coder. Stats wouldn't work otherwise. If you did, could you please send me a copy of the ones you worked on? Cheers.

On Tue, Mar 26, 2024 at 6:38 PM Kaiqin Chan <Kaiqin.chan@jcu.edu.au> wrote:

Dear Chris,

I think we should have written that we would use Cohen's kappa rather than intraclass correlation coefficient. Both are under the same family of interrater reliability statistics, but the former is for binary variables, which is the case for our variables.

If you have SPSS, Cohen's kappa can be computed under Analyze>Descriptive Statistics>Crosstabs. I believe the two free software, JASP and Jamovi, can compute Cohen's kappa as well. So if you want to try your hands at it, please feel free to try it out.

I had a look at Study 1 (not Study 2). I note the difference in coding the Participant ID column. For probe.coder 1 and probe.coder2, the Cohen's kappa was very low, at .055. This suggests that both coders need to reevaluate their coding for probe, at the very least.

I checked another a few pairs of variables, and the kappa is decent, ranging from .50 to .66, which assures me that your coders are doing the right thing, except the probe variable mentioned above. I realized that Kappa cannot be computed for some variable pairs when one of the coders had the same value, for example, the variable nontarget.low1.coder2 was "1" throughout. However, this issue isn't important.

Regardless of the kappa, eventually both coders need to complete the columns where they both come to a consensus for each variable. These columns are named "xxx.yyy.final". Would you go ahead and complete that?

Regards,
Kai Qin

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Tuesday, March 26, 2024 9:10 PM
To: Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
Cc: Adam Wang <adam.wang@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Hi Kai,

yes, we have the completed spreadsheets. There's a few caveats here.

They're the raw spreadsheets as we got them from our independent coders. I'm sending them as they are, without any of our transformations, as to not introduce any ad-hoc decisions we might have made that generate unreproducible analyses.

Study 1 has 110 participants. This is post exclusion for non-compliance and guessing the hypotheses. However, one of the coders said data on participants 10, 53, 84, and 94 was not audible and preferred not to code it. We'd rather go with her version and not include their data, as they're well over 80 participants, which was the minimal sample size.
Study 2 has 109 participants. Same for participants 33, 43, and 63, they also have to be excluded.

One more thing. Coder 1 coded all her participants in order (from 1 to 110/109 respectively using just numbers), while Coder 2 did not code them in order and used P in front of their participant numbers. So, for example, data from participant "21" from Coder 1 needs to be matched with data from participant "P21" from Coder2. We played around with doing that, but I'd rather just send the raw data files, as we have them from the coders.

If there's any other issues with the datasets, please let me know, and we'll look them over.

On Tue, Mar 26, 2024 at 11:13 AM Kaiqin Chan <Kaiqin.chan@jcu.edu.au> wrote:

> Dear Chris,
>
> Do you already have the data entered in the spreadsheet? If you do, perhaps it's easier to send me the data and then I'll compute ICC on my end and then send the results back to you?
>
> Kai Qin
>
> ---
>
> **From:** Chris Isloi <barbu.chris@gmail.com>
> **Sent:** Tuesday, March 26, 2024 4:03 PM
> **To:** Adam Wang <adam.wang@jcu.edu.au>
> **Cc:** Kaiqin Chan <Kaiqin.chan@jcu.edu.au>
> **Subject:** Re: Update on Multilab RRR

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Hi, Kaiqin, apologies for taking up your time. We've never run an ICC in our studies before and, straightforward as it may seem, we realized it can become deceptively complicated.

https://rdrr.io/cran/psych/man/ICC.html - we were browsing through here and one of our collaborators said you can't run ICC on dichotomous 0/1 data, as Study 1 requires.

Also, we don't know what the ICC cutoff is for an acceptable rating and where to go from there. Adam suggested the default SPSS version, but even that leaves questions unanswered.

Hopefully we can figure this out.

Thanks.

Chris.

On Tue, Mar 26, 2024 at 9:38 AM Adam Wang <adam.wang@jcu.edu.au> wrote:

Hi Chris

I have cc'd the stats expert of our team - Kaiqin, he should be able to give you some pointers regarding the ICC

Thanks
Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Tuesday, March 26, 2024 2:53 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Hi, Adam

following up on this. We've never worked with ICC and just now are figuring out that it may not be as easy as we thought. Do you think there's someone who could help us through it?

Thanks.

Chris.

On Thu, Mar 14, 2024 at 12:22 PM Chris Isloi <barbu.chris@gmail.com> wrote:

Got it thanks.

On Thu, Mar 14, 2024 at 12:18 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

Hi Chris

I suggest just going with the one in SPSS. None of the other labs have had this question so far so Im assuming they also just did it through SPSS

Cheers
Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Thursday, March 14, 2024 6:11 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Subject: Re: Update on Multilab RRR

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Hi, Adam, quick question, we're trying to look at our ICC between the two raters so we can estimate if there's situations where it needs to be discussed. However, I'm not sure what type of ICC you're planning on using, as the pre-print is vague about this:

https://rdrr.io/cran/psych/man/ICC.html - since there's at least 3 I know of, not sure on where to go from here. What did the other teams do?

Thanks.

Chris.

On Sun, Jan 7, 2024 at 6:09 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

Great, thanks for the update Chris!

Adam

From: Chris Isloi <barbu.chris@gmail.com>
Sent: Sunday, January 7, 2024 9:16 PM
To: Adam Wang <adam.wang@jcu.edu.au>
Subject: Re: Update on Multilab RRR

Hi Adam,

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happy new year. Yes, in the Romanian lab we'll definitely be done by May 2024.

Thanks.

Chris.

On Tue, Jan 2, 2024 at 1:29 PM Adam Wang <adam.wang@jcu.edu.au> wrote:

> Hi Everyone
>
> Hope you had a restful break. It's been another 6 months so once again I am emailing you for an update. Some of you have told me that you'd struggle to have everything completed by Dec 2023, so I have spoken to the Editor and we have agreed to extend the deadline to mid 2024 - can each of you please let me know if everything at your lab is on track for completion by May 2024?
>
> Sorry for the repeat emails if we have been in touch recently. Thanks and I look forward to hearing from you all
>
> Happy new year!
> Adam